TDNet: Bidirectional LSTM Gated Triple-Decoder Network for Remote Sensing Change Detection

Cui Zhang, Zhouyang Sha, and Hailong Wang

***Abstract*—Remote sensing change detection (CD) identifies land-use/land-cover changes by analyzing multitemporal imagery. Mainstream methods employ Siamese multilevel encoders and a single multilevel decoder: encoders extract multiscale features, while the decoder fuses them to generate change maps. However, the effective exploitation of multiscale features in existing methods remains limited, often falling short in fully unleashing the discriminative potential at each scale and in adequately achieving synergistic cross-scale fusion. To address this limitation, we propose the bidirectional long short-term memory (LSTM) gated triple-decoder network (TDNet), which introduces two innovations: 1) triple-decoder architecture: we abandon the single-decoder paradigm and construct three cascaded decoder branches of varying depths (two-four layers). Each branch specializes in processing features at its corresponding level—shallow, middle, or deep—fully unleashing the discriminative potential at every scale and 2) bidirectional LSTM-based multiscale feature interaction module (BLMIM): leveraging LSTM's gating mechanism, BLMIM extends the bidirectional LSTM to handle multiscale feature interactions: it adaptively retains upstream features via the forget gate, filters current-layer information via the input gate, and controls contributions to subsequent layers via the output gate. This yields interpretable multiscale interactions and, to the authors' knowledge, constitutes the first adaptation of LSTM to supervised CD. Comprehensive experiments on LEVIR-CD, CDD, and WHU-CD demonstrate that TDNet surpasses state-of-the-art methods, achieving $F1$ improvements of 0.60%, 0.49%, and 0.50%, respectively.**

***Index Terms*—Change detection (CD), deep learning, Mamba, remote sensing.**

I. Introduction

REMOTE sensing change detection (CD) aims to accurately identify the dynamic evolution of land use/land cover through comparative analysis of multitemporal imagery of identical geographical areas. This technology has been extensively applied in agricultural monitoring, environmental conservation, disaster assessment, and urban planning [1], [2], [3]. To support these diverse applications, significant research efforts have been dedicated to developing powerful CD models.

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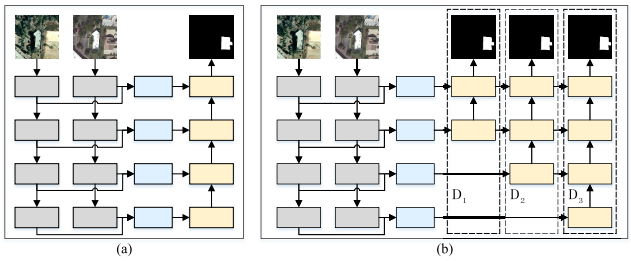


Fig. 1. (a) Single-decoder architecture. (b) Triple-decoder architecture. Each gray box represents one encoder level, each yellow box represents one level within a decoder branch, and each blue box denotes a feature fusion or difference feature extraction module. D_1, D_2, and D_3 denote the three decoder branches in the triple-decoder architecture.

Since deep learning was introduced into CD, convolutional neural network (CNN) [4], [5] and Transformer [6], [7] have successively taken leading roles in methodological advances. CNN-based CD methods [8], [9], [10], [11], [12], [13] leverage local connectivity and weight-sharing mechanisms to achieve fine-grained spatial feature extraction while maintaining computational efficiency. Transformer-based CD methods [14], [15], [16] employ global self-attention to capture long-range dependencies, significantly enhancing large-scale contextual modeling capabilities. Recently, Mamba-based CD approaches [17], [18] have emerged, utilizing selective scan space state sequential models (S6) [19] to establish global correlations with linear time complexity, further advancing long-range feature extraction efficiency.

Most existing CD methods employ a Siamese encoder–single decoder architecture [as shown in Fig. 1(a)]: multilevel Siamese encoders extract multiscale features from bitemporal images, while a single multilevel decoder utilizes these features to generate change maps. However, this design suffers from two limitations: 1) it fails to fully unleash the discriminative potential at each feature scale and 2) there is room for improvement in achieving effective interaction between different scales. Consequently, the multiscale features are underexploited, which compromises the precise identification of change areas and constrains the overall detection accuracy.

To overcome the aforementioned limitation, we propose the bidirectional long short-term memory (LSTM) gated triple-decoder network (TDNet). Departing from conventional single-decoder designs, TDNet pioneers a triple-decoder architecture [as shown in Fig. 1(b)].

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1) Decoder D_1 (two layers) processes features from encoder layers 1 and 2.
2) Decoder D_2 (three layers) processes features from encoder layer 3 together with the one–two-layer features from D_1.
3) Decoder D_3 (four layers) processes features from encoder layer 4 along with the one–three-layer features from D_2.

This cascade architecture progressively calibrates and refines features at each level, enabling D_3 to generate more discriminative change representations and a more accurate change map.

Furthermore, we introduce a bidirectional LSTM-based multiscale feature interaction module (BLMIM) to enhance cross-layer feature complementarity. BLMIM leverages LSTM [20], [21], [22] gating mechanisms: forget gates modulate the retention of upstream-level features. Input gates modulate the selection of current-level features. Output gates modulate the propagation of output features to downstream layers. This module enhances the efficacy and interpretability of multiscale feature interaction. According to the authors' knowledge, this is the first work to adapt LSTM's gating strategy for supervised CD.

The choice of encoder backbone is critical, as it fundamentally governs the quality of feature representations in CD models. Contemporary methods typically adopt backbones such as ResNet [23], PVT [24], VMamba [25], or MobileSAM [26] as their encoders. To validate the generalization capability of TDNet, we construct its encoder using each of these heterogeneous backbones and conduct comprehensive experimental evaluations.

The core contributions of this article can be summarized as follows.

1) *Proposal of the Bidirectional LSTM Gated TDNet:* Breaking through the limitations of traditional single-decoder architectures, we pioneer a triple-decoder architecture. This design employs a feature cascade refinement mechanism to fully exploit the discriminative potential of features across all levels, significantly enhancing the model's accuracy.
2) *Design of the Bidirectional LSTM-Based Multiscale Feature Interaction Module (BLMIM):* We are the first to utilize LSTM's gating mechanisms (forget/input/output gates) for multiscale feature interaction. This enables adaptive modulation of cross-scale feature exchange, enhancing both the effectiveness and interpretability of multiscale feature fusion.
3) *Demonstration of Significant Performance Gains and Validation of Generalizability:* On three public datasets, $F1$ increased by 0.6%, 0.49%, and 0.5%, respectively (compared to state of the art). Both the triple-decoder architecture and BLMIM demonstrate consistent performance gains across heterogeneous backbones, including ResNet, PVT, VMamba, and MobileSAM, systematically validating the method's effectiveness, generalizability, and backbone-agnostic property.

The rest of this article is structured as follows. Section II covers related work. Section III describes the proposed methodology. Section IV details the experiments and results. Section V provides the conclusion.

II. Related Work

A. CNN-Based CD Methods

Since the introduction of AlexNet [5], CNN has rapidly become the dominant architecture in computer vision, giving rise to numerous classical methodologies for CD tasks. Early representative works include: Zhan et al. [8] established the fundamental paradigm for dual-temporal feature extraction through a Siamese dual-branch encoder architecture. Caye Daudt et al. [9] developed a CD framework based on fully convolutional networks and implemented an end-to-end training strategy.

To enhance the performance of CNN-based CD methods, researchers have proposed several improvements: Zhang et al. [10] employed dilated convolutions [27] to expand receptive fields while integrating triplet loss [28] to address data imbalance. Chen and Shi [11] and Chen et al. [12] constructed CD networks using ResNet [23] architectures, respectively, incorporating a pyramid spatial–temporal attention module and a dual attention module to optimize feature representation. Fang et al. [13] and Li et al. [29] adopted densely connected structures [30] to strengthen feature reuse, respectively, combining an ensemble channel attention module and a hybrid attention module to enhance feature robustness. Liu et al. [31] proposed a multistage difference enhancement module and a segmentation-enhanced loss function, which significantly improved the detection of subtle changes in large-size remote sensing images. Miao et al. [32] introduced an innovative generative adversarial network to effectively distinguish pseudo changes caused by seasonal variations and actual land surface changes. Xing et al. [33] proposed an edge enhancement module that extracts and enhances high-frequency features to improve the representation of edge details, thereby achieving finer grained detection of change edges.

Despite these enhancements, CNN-based CD methods still fall short in global modeling, constraining their ultimate accuracy.

B. Transformer-Based CD Methods

The global modeling capability of Transformer [6] has led to their rapid rise in the vision domain. The success of ViT [7] has propelled their application in remote sensing CD.

Chen et al. [14] were the first to introduce Transformer into CD. They added a Transformer encoder–decoder after a ResNet18 backbone network to extract global features. Li et al. [15] and Song et al. [34] also adopted a similar paradigm, embedding Transformer modules after CNN backbone networks. Bandara and Patel [16] proposed a CD network that integrates a Transformer encoder with a multilayer perceptron (MLP) decoder. Feng et al. [35] designed a network with a dual-branch encoder structure, where one branch is based on ResNet18 and the other on PVTv2 [36]. Tang et al. [37] similarly employed a dual-branch encoder structure. Differing from others, to better model objects with diverse shapes, they utilized deformable convolution [38] and deformable attention

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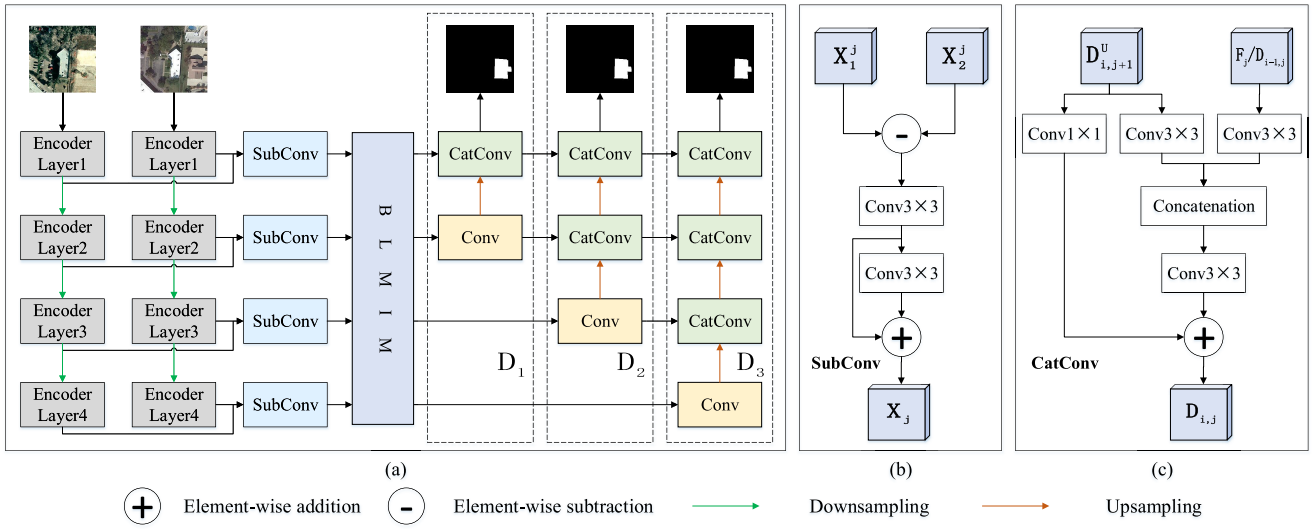


Fig. 2. Architecture of the bidirectional LSTM gated TDNet. (a) Overall framework. (b) SubConv module. (c) CatConv module.

Transformer [39]. Zhang et al. [40] proposed asymmetric cross-attention to enhance the global perception capability of features at each level. Han et al. [41] employed hybrid self-attention and cross-attention to capture global and cross-scale information. However, the quadratic computational complexity of Transformer constrains the potential application of Transformer-based methods.

C. Mamba-Based CD Methods

Mamba [19] achieves global modeling with linear computational complexity through state-space models, providing a new approach for CD. VMamba [25] introduced the cross-scan module, which employs a four-way scanning strategy to address the directional sensitivity of visual data, laying the foundation for subsequent research. In the field of CD, Chen et al. [17] proposed ChangeMamba, pioneering the application of Mamba to CD tasks; Zhang et al. [18] proposed CDMamba, designing a ConvMamba module that utilizes CNN to extract local features and Mamba to extract global features. Current Mamba-based methods still face a key challenge: their sequential processing mechanism has difficulty adequately preserving the spatial structural information of image data, which requires further research and improvement.

III. Methodology

In this article, we propose a bidirectional LSTM gated TDNet, whose architecture is illustrated in Fig. 2. Consistent with most CD networks [9], [13], [42], TDNet employs a Siamese-encoder architecture. Its distinctive innovation lies in the triple-decoder architecture and a BLMIM embedded between the encoder and decoder.

A. Encoder

As illustrated in Fig. 2(a), TDNet's encoder adopts a four-level pyramid architecture. The network accepts a bitemporal image pair (I_1, I_2) as input, with each image processed through dedicated encoder branches to extract multiscale features represented as $\{X_i^1, X_i^2, X_i^3, X_i^4\}_{i=1}^2$ at four hierarchical levels, where i denotes the temporal index.

For features at each hierarchical level j (where $j \in \{1, 2, 3, 4\}$), the corresponding temporal features X_1^j and X_2^j are input to a SubConv module to extract preliminary change features X_j. The specific structure of the SubConv module is shown in Fig. 2(b), which mainly consists of an elementwise subtraction operation and two convolution operations.

As the key component of CD networks, the encoder design critically determines model performance. Prevailing approaches typically employ backbone networks with multiscale feature extraction capabilities—such as ResNet, PVT, VMamba, and MobileSAM—to construct encoders. Given that CD datasets are generally limited in scale, the encoders in these approaches are initialized using ImageNet pretrained weights to enhance the generalization capability of feature representations.

In this article, we do not rely on a single backbone but instead construct TDNet with each of the aforementioned backbones. The objective of this manner is to demonstrate the stability and generalizability of the triple-decoder and BLMIM across different backbones through comprehensive experiments.

B. BLMIM

The BLMIM enables cross-scale adaptive feature fusion. Multiscale features extracted by the encoder exhibit asymmetric limitations: The multiscale features extracted by the encoder exhibit the following limitations.

1) Shallow features (higher spatial resolution) excel at capturing fine-grained details but lack holistic contextual awareness.

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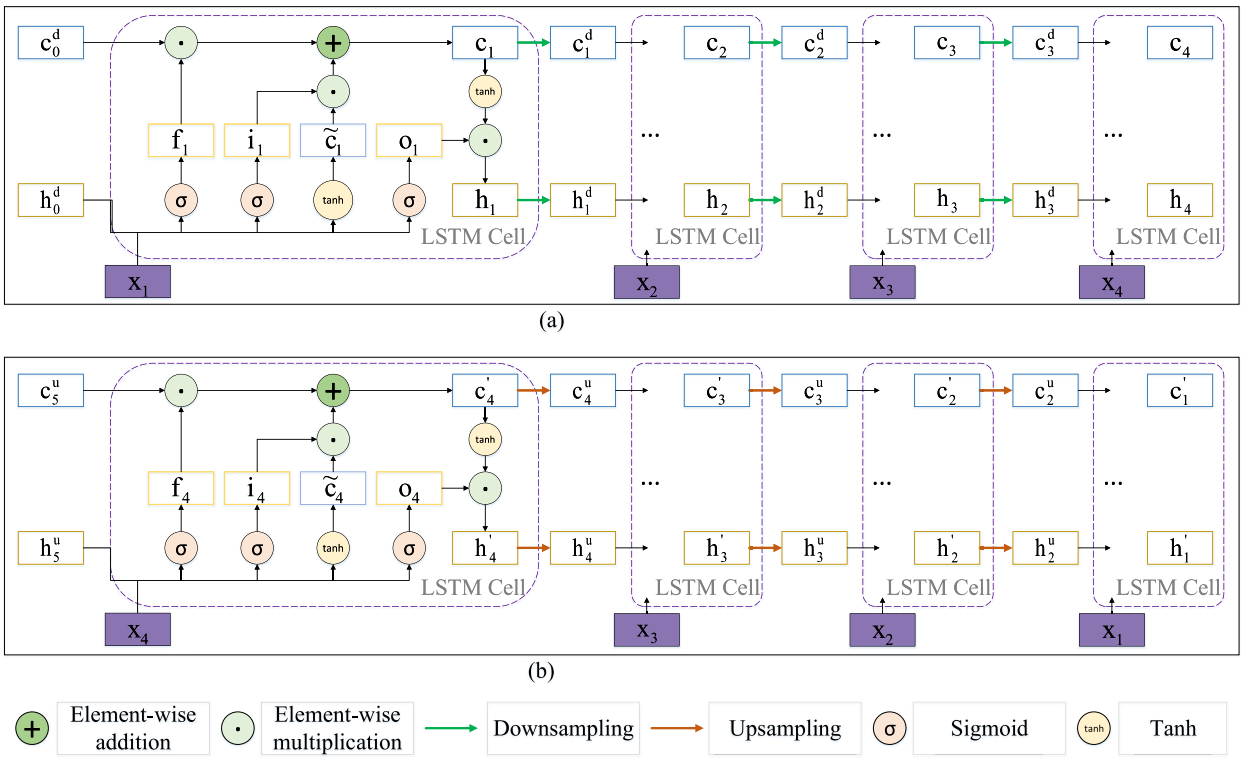


Fig. 3. Bidirectional LSTM-based multiscale interaction module. (a) Forward calculation. (b) Backward calculation.

2) Deep features (lower spatial resolution) encode global semantics but suffer from detail degradation due to progressive downsampling.

BLMIM addresses this through LSTM gating mechanisms that regulate interscale information flow. Features at each scale dynamically incorporate complementary information from other levels, yielding more comprehensive and robust representations.

Specifically, the LSTM gating mechanism processes information iteratively, computing three gates at each time step.

1) Forget gate regulates what information to disengage from the previous cell state.
2) Input gate modulates how much of the candidate state enters the memory cell.
3) Output gate regulates what portion of the current cell state to output to external states.

The dual requirements in multiscale feature fusion—integrating complementary information from other scales and preserving effective information at the current scale—directly align with the core philosophy of LSTM's gating mechanism. This alignment motivates our adoption of its principles for achieving cross-scale interaction.

We implement a bidirectional LSTM framework that processes multiscale features in two directions: forward computation sequentially integrating features from Scale 1 (shallow) through Scale 4 (deep), while reverse computation sequentially integrates features from Scale 4 (deep) through Scale 1 (shallow). However, significant resolution heterogeneity prevents direct application. During LSTM computation at each scale, we adjust the resolution of features transferred from adjacent scales—downsampling when propagating to coarser resolutions or upsampling when propagating to finer resolutions—to ensure resolution consistency for LSTM operations.

Fig. 3 details the BLMIM for multiscale feature interaction. This process mainly consists of two steps: forward computation and reverse computation.

1) Forward Computation: It processes multiscale features sequentially from fine to coarse scales ($i = 1 \rightarrow 2 \rightarrow 3 \rightarrow 4$): At each scale i, the BLMIM proceeds as follows.

a) State downsampling:

$$i = 1: \quad c_0^d = \mathbf{0}, \quad h_0^d = \mathbf{0} \in \mathbb{R}^{\frac{H}{4} \times \frac{W}{4} \times C} \tag{1}$$

$$i > 1: \quad \left(c_{i-1}^d, h_{i-1}^d\right) = \text{Downsample}\left(c_{i-1}, h_{i-1}\right) \tag{2}$$

where Downsample represents the downsampling operation, decreasing spatial resolution from $H/(2^i) \times W/(2^i)$ to $H/(2^{i+1}) \times W/(2^{i+1})$. H and W represent the height and width of the input image, respectively. C represents the channel number. c_i and h_i denote the cell state and hidden state at the ith layer, respectively.

b) Gated state update:

$$c_i, h_i = \text{LSTM}_{\text{cell}}\left(x_i, \left(c_{i-1}^d, h_{i-1}^d\right); \Theta\right). \tag{3}$$

The specific calculation of $\text{LSTM}_{\text{cell}}$ is as follows.

1) Gating and candidate state computation

$$f_i = \sigma\left(\mathbf{W}_f x_i + \mathbf{U}_f h_{i-1}^d + \mathbf{b}_f\right) \tag{4}$$

$$i_i = \sigma\left(\mathbf{W}_i x_i + \mathbf{U}_i h_{i-1}^d + \mathbf{b}_i\right) \tag{5}$$

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$$o_i = \sigma\left(\mathbf{W}_o x_i + \mathbf{U}_o h_{i-1}^d + \mathbf{b}_o\right) \tag{6}$$

$$\tilde{c}_i = \tanh\left(\mathbf{W}_c x_i + \mathbf{U}_c h_{i-1}^d + \mathbf{b}_c\right) \tag{7}$$

where f_i, i_i, and o_i denote the forget gate, input gate, and output gate at the ith layer, respectively. $\tilde{c}_i$ denotes the candidate state. $\mathbf{W}_*, \mathbf{U}_* \in \mathbb{R}^{C\times C}$ and $\mathbf{b}_* \in \mathbb{R}^C$ in Θ are the learnable parameters.

2) State update

$$c_i = f_i \odot c_{i-1}^d + i_i \odot \tilde{c}_i \tag{8}$$

$$h_i = o_i \odot \tanh\left(c_i\right). \tag{9}$$

Through forward computation, we sequentially obtain the hidden states $\{h_1, h_2, h_3, h_4\}$.

2) Reverse Computation: It processes multiscale features sequentially from coarse to fine scales ($i = 4 \to 3 \to 2 \to 1$). At each scale i, the BLMIM proceeds as follows.

a) State upsampling:

$$i = 4: \quad c_5^u = \mathbf{0}, \quad h_5^u = \mathbf{0} \in \mathbb{R}^{\frac{H}{32}\times\frac{W}{32}\times C} \tag{10}$$

$$i < 4: \quad \left(c_{i+1}^u, h_{i+1}^u\right) = \text{Upsample}\left(c_{i+1}', h_{i+1}'\right) \tag{11}$$

where Upsample denotes the upsampling operation, increasing spatial resolution from $H/(2^{i+2}) \times W/(2^{i+2})$ to $H/(2^{i+1}) \times W/(2^{i+1})$.

b) Gated state update:

$$c_i', h_i' = \text{LSTM}_{\text{cell}}\left(x_i, \left(c_{i+1}^u, h_{i+1}^u\right); \Theta'\right). \tag{12}$$

Note: Parameters $\mathbf{W}_*'$, $\mathbf{U}_*'$, and $\mathbf{b}_*'$ in Θ' are distinct from their forward computation counterparts.

Through reverse computation, we sequentially obtain the hidden states $\{h_4', h_3', h_2', h_1'\}$.

Following forward and backward computation, BLMIM aggregates x_i with h_i and h_i' to obtain enhanced features F_i at each scale. The aggregation process is defined as

$$F_i = \text{MLP}\left(\text{Concat}\left(h_i, h_i'\right)\right) + x_i \tag{13}$$

where Concat denotes the channelwise concatenation operation, MLP represents a multilayer perceptron primarily composed of fully connected layers, and x_i is the original encoder feature at scale i.

C. Triple-Decoder

To fully exploit multiscale features, TDNet incorporates a triple-decoder architecture with three branches: D_1, D_2, and D_3, designed under a progressive feature refinement paradigm, as illustrated in Fig. 2(a). This architecture diverges from the single decoder of U-Net [43] and the nested, dense skip connections of UNet++ [44]. Instead, it employs a novel cascade architecture where each decoder refines the features from its predecessor to progressively enhance the change representation.

1) Decoder D_1: It consists of two layers. Its input is $\{F_2, F_1\}$ from BLMIM

$$D_{1,2} = \text{Conv}\left(F_2\right) \tag{14}$$

$$D_{1,2}^u = \text{Upsample}\left(D_{1,2}\right) \tag{15}$$

$$D_{1,1} = \text{CatConv}\left(D_{1,2}^u, F_1\right). \tag{16}$$

The specific structure of the CatConv module is shown in Fig. 2(c), primarily comprising convolution operations.

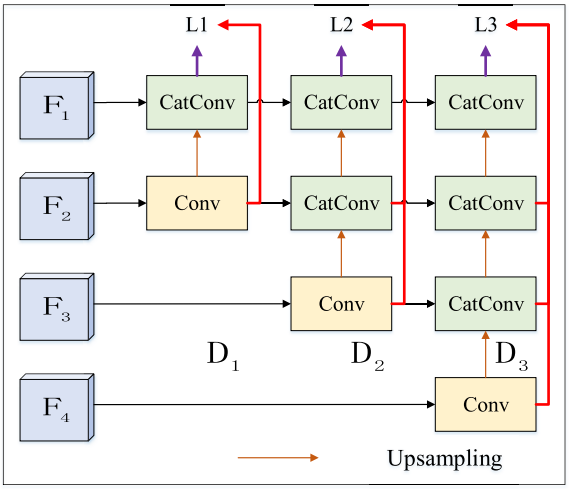


Fig. 4. Comprehensive deep supervision strategy in TDNet.

2) Decoder D_2: It contains three layers, with inputs being F_3 from BLMIM, and $\{D_{1,2}, D_{1,1}\}$ from Decoder D_1

$$D_{2,3} = \text{Conv}\left(F_3\right) \tag{17}$$

$$D_{2,3}^u = \text{Upsample}\left(D_{2,3}\right) \tag{18}$$

$$D_{2,2} = \text{CatConv}\left(D_{2,3}^u, D_{1,2}\right) \tag{19}$$

$$D_{2,2}^u = \text{Upsample}\left(D_{2,2}\right) \tag{20}$$

$$D_{2,1} = \text{CatConv}\left(D_{2,2}^u, D_{1,1}\right). \tag{21}$$

3) Decoder D_3: It contains four layers, with inputs being F_4 from BLMIM, and $\{D_{2,3}, D_{2,2}, D_{2,1}\}$ from Decoder D_2

$$D_{3,4} = \text{Conv}\left(F_4\right) \tag{22}$$

$$D_{3,4}^u = \text{Upsample}\left(D_{3,4}\right) \tag{23}$$

$$D_{3,3} = \text{CatConv}\left(D_{3,4}^u, D_{2,3}\right) \tag{24}$$

$$D_{3,3}^u = \text{Upsample}\left(D_{3,3}\right) \tag{25}$$

$$D_{3,2} = \text{CatConv}\left(D_{3,3}^u, D_{2,2}\right) \tag{26}$$

$$D_{3,2}^u = \text{Upsample}\left(D_{3,2}\right) \tag{27}$$

$$D_{3,1} = \text{CatConv}\left(D_{3,2}^u, D_{2,1}\right). \tag{28}$$

4) Comprehensive Deep Supervision: We introduce a comprehensive deep supervision strategy to directly steer the progressive refinement process, as illustrated in Fig. 4. Similar to [44], we apply supervision to the final output features of the three decoder branches ($D_{1,1}, D_{2,1}$, and $D_{3,1}$), indicated by the purple lines in Fig. 4. In addition, we also impose supervision on key intermediate features during the decoding process ($D_{1,2}, D_{2,3}, D_{2,2}, D_{3,4}, D_{3,3}$, and $D_{3,2}$), represented by the red lines in Fig. 4. This design ensures that the features evolve in the correct direction at each step of the cascaded transmission, thereby guaranteeing the effectiveness of the entire progressive refinement mechanism. All supervised features are passed through the change prediction head to generate corresponding change maps, which subsequently contribute to the computation of the final loss function.

5) Change Prediction Head: $D_{i,j}$ is processed through the change prediction head to obtain the change map $P_{i,j}$,

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where $i \in [1,3]$ and $j \in [1, i+1]$. The specific computational procedure of the change prediction head is defined as

$$P_{i,j} = \text{Upsample}\left(\text{Linear}\left(D_{i,j}\right)\right) \tag{29}$$

where Linear denotes a fully connected layer. Notably, $P_{3,1}$ serves as the final change map of TDNet.

D. Loss Function

Consistent with [45], we employ a hybrid loss function combining BCE loss and Dice loss [46]

$$L(P,G) = L_{\text{BCE}}(P,G) + L_{\text{Dice}}(P,G) \tag{30}$$

$$L_{\text{BCE}}(P,G) = -\frac{1}{N}\sum_{i=1}^{N}\left[g_i \log(p_i) + (1-g_i)\log(1-p_i)\right] \tag{31}$$

$$L_{\text{Dice}}(P,G) = 1 - \frac{2\sum_{i=1}^{N} p_i g_i}{\sum_{i=1}^{N} p_i + \sum_{i=1}^{N} g_i + \epsilon} \tag{32}$$

where

L	hybrid loss function;
P	predicted change map of dimensions $H \times W$;
G	ground truth of dimensions $H \times W$;
$N = H \times W$	total number of pixels;
i	pixel index ranging over all spatial positions ($i \in \{1,2,\ldots,N\}$);
p_i	predicted change probability at pixel i ($p_i \in [0,1]$);
g_i	ground-truth binary label at pixel i ($g_i \in \{0,1\}$);
$\epsilon = 10^{-5}$	smoothing constant preventing division by zero.

Given that TDNet comprises three decoder branches, which generate multiple change maps—$\{P_{1,1}, P_{1,2}\}$ from the first branch, $\{P_{2,1}, P_{2,2}, P_{2,3}\}$ from the second, and $\{P_{3,1}, P_{3,2}, P_{3,3}, P_{3,4}\}$ from the third—the total loss function is calculated as follows:

$$L_{\text{total}} = \sum_{i=1}^{3} \alpha_i L_i \tag{33}$$

$$L_i = \sum_{j=1}^{i+1} L\left(P_{i,j}, G\right) \tag{34}$$

where α_i represents the weight coefficient of L_i, and all α_i are set to 1 by default.

IV. Experiments

A. Datasets

To comprehensively evaluate the generalization capability and robustness of TDNet, experiments are conducted on three widely used public CD benchmarks.

LEVIR-CD Dataset [11]: It comprises 637 pairs of satellite images (1024 × 1024 pixels, 0.5-m spatial resolution) documenting urban building construction and demolition. Following standard protocol [11], the dataset is partitioned into the following:

1) 445 image pairs for training;
2) 128 image pairs for testing;
3) 64 image pairs for validation.

WHU-CD Dataset [47]: It consists of one aerial image pair (32507 × 15354 pixels, 0.075-m spatial resolution) capturing postearthquake urban reconstruction. We process this dataset by cropping the original images into 9860 nonoverlapping patches. The resulting patches are distributed as follows:

1) 7918 patches for training;
2) 987 patches for testing;
3) 955 patches for validation.

CDD Dataset [48]: It contains 11 pairs of satellite images (4725 × 2200 to 1900 × 1000 pixels, 0.03–1-m resolution) featuring diverse changes including buildings, roads, and vehicles. Using the official partitioning [48], images are segmented into 16 000 patches of 256 × 256 pixels:

1) 10 000 patches for training;
2) 3000 patches for testing;
3) 3000 patches for validation.

B. Evaluation Criteria

To comprehensively quantify model performance, five rigorous evaluation metrics are employed.

1) Precision
$$P = \frac{\text{TP}}{\text{TP}+\text{FP}} \tag{35}$$
measures the proportion of correctly identified changes among all detected changes.
2) Recall
$$R = \frac{\text{TP}}{\text{TP}+\text{FN}} \tag{36}$$
evaluates the completeness of true CD.
3) $F1$
$$F1 = \frac{2 \times P \times R}{P+R} \tag{37}$$
harmonic mean of precision and recall, serving as the core comprehensive performance indicator.
4) Intersection over union (IoU)
$$\text{IoU} = \frac{\text{TP}}{\text{TP}+\text{FP}+\text{FN}} \tag{38}$$
directly measures spatial overlap between predicted and actual change regions.
5) Overall accuracy (OA)
$$\text{OA} = \frac{\text{TP}+\text{TN}}{\text{TP}+\text{FP}+\text{FN}+\text{TN}} \tag{39}$$
assesses global classification consistency across all pixels.

where

TP =	true positives (correctly detected changed pixels);
TN =	true negatives (correctly identified unchanged pixels);
FP =	false positives (unchanged pixels misclassified as changed);
FN =	false negatives (changed pixels missed by detection).

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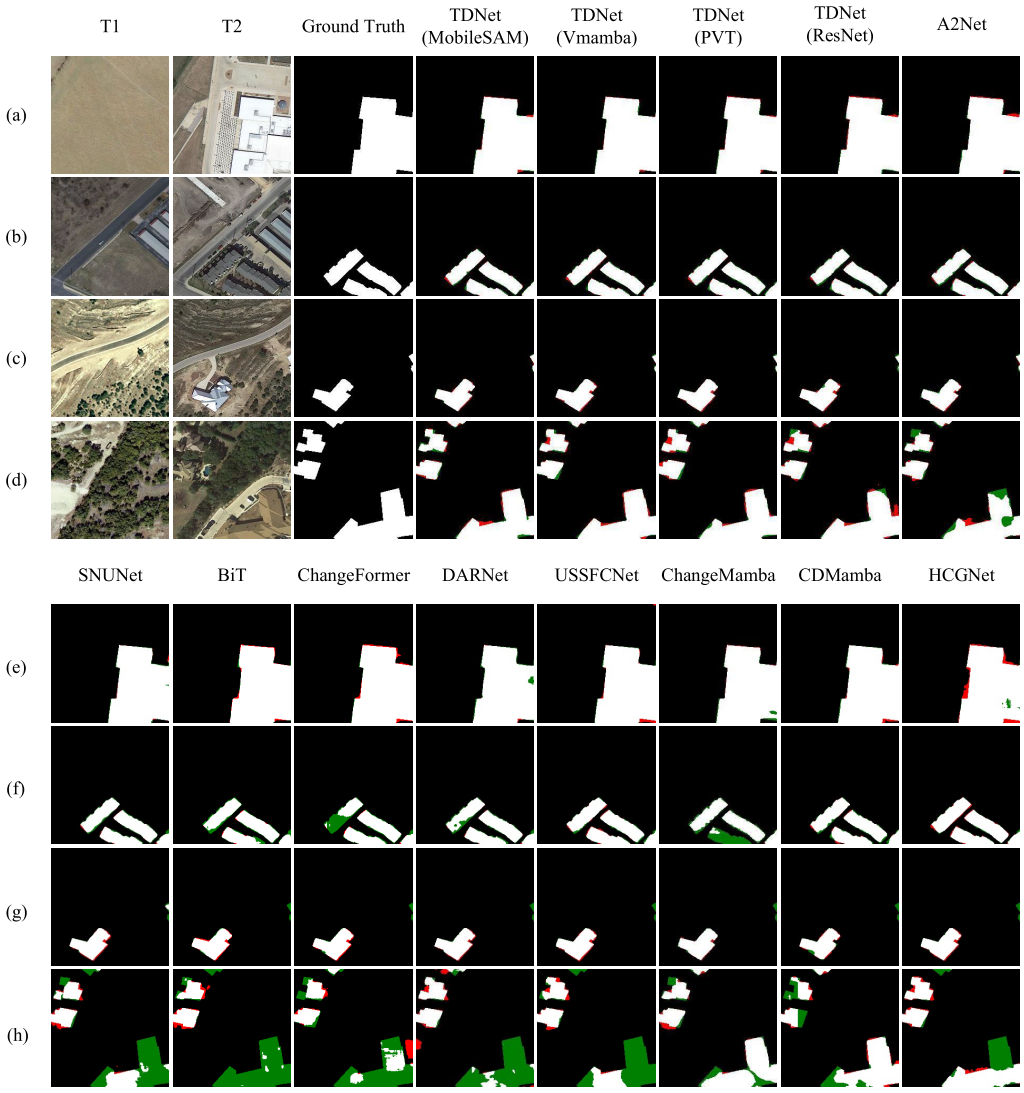


Fig. 5. Visualization results of several CD methods on LEVIR-CD test set. Green regions indicate missed changes (false negatives), and red regions denote false alarms (false positives). (a)–(d) Present four sample sets selected from the test set, along with the corresponding change maps generated by our proposed method and A2Net, (e)–(h) are a direct horizontal extension of (a)–(d), showcasing the change maps produced by various other comparative methods.

C. Comparative Experiments

In comparative experiments, we benchmarked TDNet against several state-of-the-art CD methods, including SNUNet [13], BiT [14], ChangeFormer [16], DARNet [29], SwinSUNet [42], USSFCNet [49], A2Net [45], HCGNet [50], CBSANet [51], GCD-DDPM [52], ChangeMamba [17], and CDMamba [18].

1) Comparative Experiments on LEVIR-CD Dataset: On the LEVIR-CD test set, TDNet outperforms existing methods across multiple backbone networks. As shown in Table I, with the MobileSAM backbone, TDNet achieves top-ranked results in three metrics: 92.21% $F1$, 99.21% OA, and 85.55% IOU. Compared to HCGNet ($F1$ of 91.61% and IoU of 84.52%), it improves $F1$ by 0.6% and IoU by 1.03%. With the VMamba backbone, TDNet also attains second-best $F1$ at 92.05%, comprehensively surpassing recent Mamba baselines like ChangeMamba and CDMamba, demonstrating the method's robustness.

TABLE I
COMPARISON RESULTS ON LEVIR-CD TEST SET. BEST RESULTS SHOWN IN RED AND SECOND BEST IN BLUE

Method	Pre. (%)	Rec. (%)	F1 (%)	OA (%)	IoU(%)
SNUNet [13]	91.47	89.13	90.28	99.02	82.29
BIT [14]	92.34	87.35	89.78	99.00	81.45
ChangeFormer [16]	91.84	87.45	89.59	98.98	81.15
DARNet [29]	91.94	88.96	90.42	99.05	82.52
SwinSUNet [42]	87.94	90.25	89.08	98.87	80.31
USSFCNet [49]	88.81	90.97	89.88	98.98	81.61
A2Net [45]	92.29	90.76	91.52	99.14	84.36
HCGNet [50]	90.82	92.42	91.61	99.14	84.52
CBSANet [51]	92.47	90.33	91.39	99.13	84.14
GCD-DDPM [52]	90.68	91.24	90.96	99.14	83.65
ChangeMamba [17]	95.11	86.01	90.33	99.06	82.36
CDMamba [18]	91.41	90.05	90.72	99.06	83.02
TDNet(ResNet)	92.62	90.75	91.67	99.16	84.48
TDNet(PVT)	92.66	90.82	91.73	99.17	84.72
TDNet(VMamba)	92.78	91.32	92.05	99.20	85.26
TDNet(MobileSAM)	93.21	91.23	92.21	99.21	85.55

Fig. 5 shows CD results predicted by multiple methods. The four test images contain buildings with distinct morphological characteristics: 1) a single large-scale building; 2) multiple closely connected buildings; 3) a building with vibrant color;

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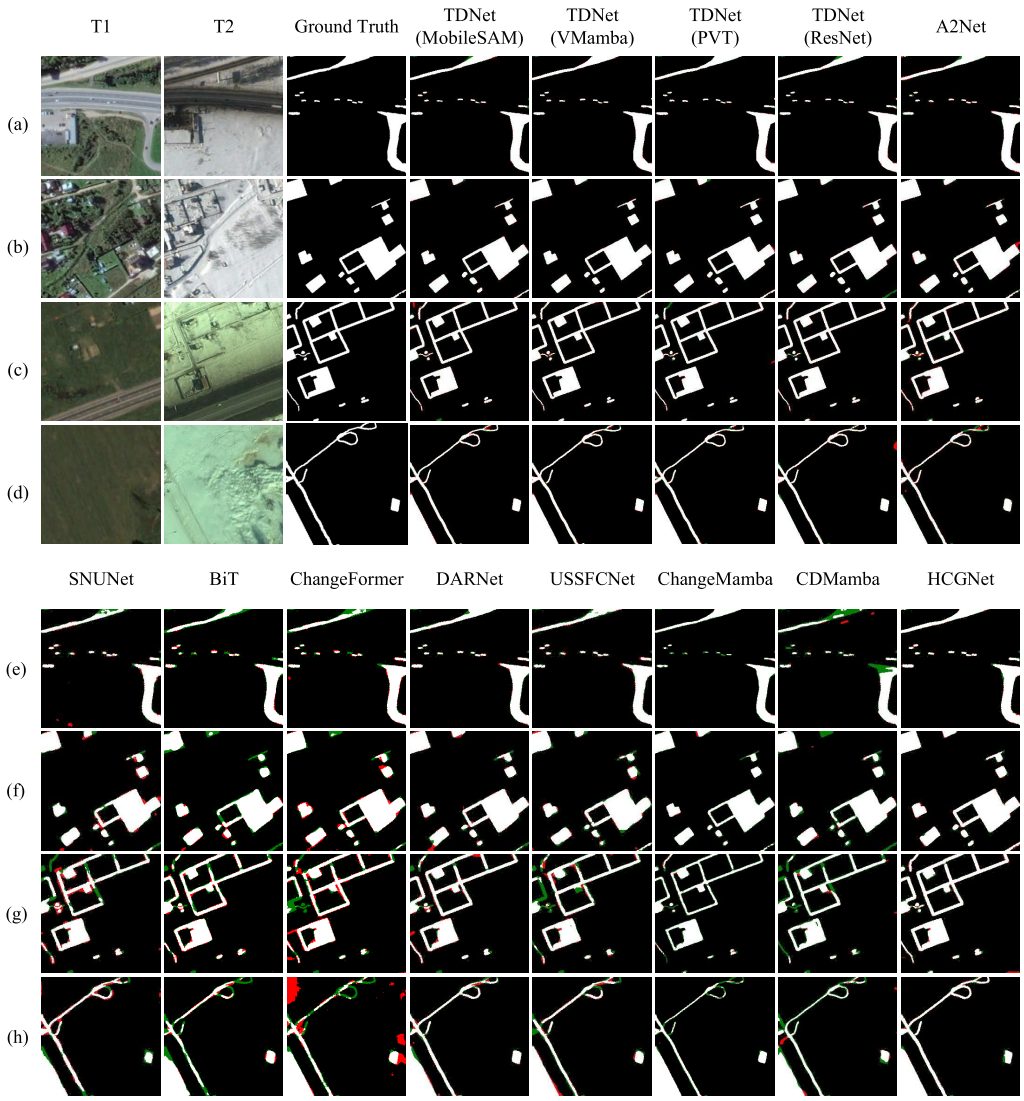


Fig. 6. Visualization results of several CD methods on the CDD test set. Green regions indicate missed changes (false negatives), and red regions denote false alarms (false positives). (a)–(d) Present four sample sets selected from the test set, along with the corresponding change maps generated by our proposed method and A2Net, (e)–(h) are a direct horizontal extension of (a)–(d), showcasing the change maps produced by various other comparative methods.

and 4) a building with dull color that is difficult to distinguish from the surrounding terrain. Visual analysis confirms that TDNet consistently produces visually accurate change maps across all four backbone configurations.

2) Comparative Experiments on CDD Dataset: On the CDD test set, TDNet achieved leading performance across all four backbone networks, as shown in Table II. With the VMamba backbone, it attained state-of-the-art results in all five metrics (98.77% precision, 98.76% recall, 98.76% $F1$, 99.69% OA, and 97.56% IoU), improving $F1$ by 0.49% and IoU by 0.96% over prior best HCGNet (98.27% $F1$ and 96.60% IoU). All backbone configurations outperformed existing methods, demonstrating consistent robustness.

Fig. 6 presents visual comparisons of CD results from multiple methods on the CDD test set. Visual inspection reveals that TDNet with VMamba backbone yields the most precise change maps, demonstrating the closest alignment with ground truth. The other three backbone configurations of TDNet also produce considerably accurate results, outperforming all comparative methods.

TABLE II
COMPARISON RESULTS ON CDD TEST SET. BEST RESULTS SHOWN IN RED AND SECOND BEST IN BLUE

Method	Pre. (%)	Rec. (%)	F1 (%)	OA (%)	IoU(%)
SNUNet [13]	95.83	93.49	94.65	98.63	89.84
BIT [14]	96.68	93.92	95.28	98.80	90.98
ChangeFormer [16]	95.44	94.01	94.72	98.65	89.97
DARNet [29]	97.07	95.57	96.31	99.05	92.89
SwinSUNet [42]	95.73	92.30	93.99	98.47	88.65
USSFCNet [49]	94.64	96.08	95.36	98.81	91.12
A2Net [45]	98.10	97.96	98.03	99.51	96.14
HCGNet [50]	98.76	97.79	98.27	99.56	96.60
CBSANet [51]	96.99	97.44	97.21	99.34	94.58
GCD-DDPM [52]	94.76	95.10	94.93	98.87	90.56
ChangeMamba [17]	97.16	97.71	97.43	99.39	94.99
CDMamba [18]	96.71	94.90	95.79	98.97	91.92
TDNet(ResNet)	98.61	98.45	98.53	99.64	97.11
TDNet(PVT)	98.73	98.47	98.60	99.65	97.23
TDNet(VMamba)	98.77	98.76	98.76	99.69	97.56
TDNet(MobileSAM)	98.62	98.49	98.56	99.64	97.15

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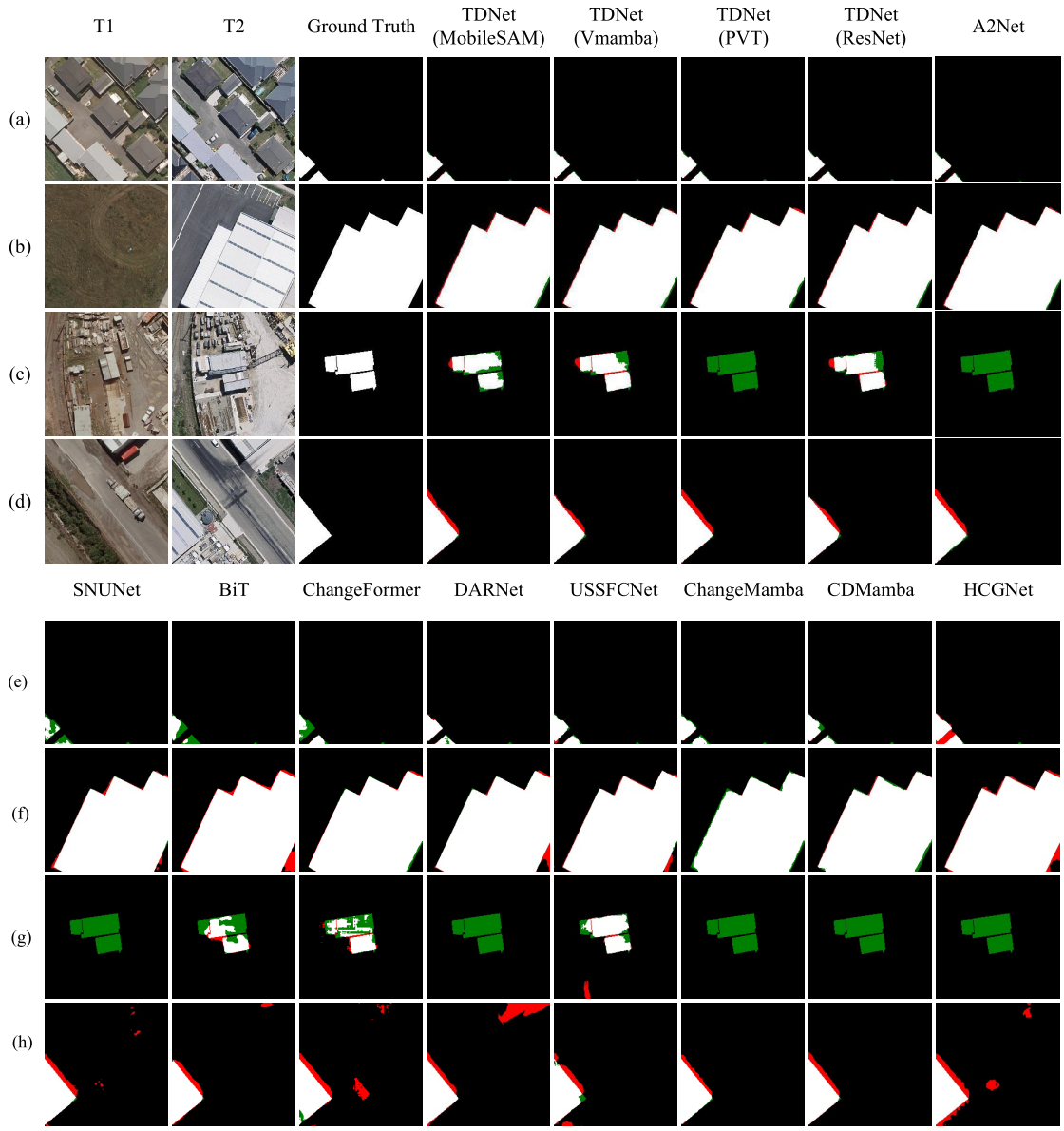


Fig. 7. Visualization results of several CD methods on the WHU-CD test set. Green regions indicate missed changes (false negatives), and red regions denote false alarms (false positives). (a)–(d) Present four sample sets selected from the test set, along with the corresponding change maps generated by our proposed method and A2Net, (e)–(h) are a direct horizontal extension of (a)–(d), showcasing the change maps produced by various other comparative methods.

TABLE III

COMPARISON RESULTS ON WHU-CD TEST SET. BEST RESULTS SHOWN IN RED AND SECOND BEST IN BLUE

Method	Pre. (%)	Rec. (%)	F1 (%)	OA (%)	IoU(%)
SNUNet [13]	91.10	81.98	86.30	98.74	75.90
BIT [14]	89.09	90.42	89.75	99.00	81.41
ChangeFormer [16]	92.40	87.05	89.65	99.02	81.23
DARNet [29]	89.74	91.67	90.70	99.09	82.98
SwinSUNet [42]	95.03	92.61	93.81	99.41	88.34
USSFCNet [49]	90.04	95.00	92.45	99.29	85.96
A2Net [45]	97.24	93.31	95.23	99.55	90.90
HCGNet [50]	95.84	91.64	93.69	99.40	88.14
CBSANet [51]	93.93	91.15	92.52	99.31	86.08
GCD-DDPM [52]	94.76	95.10	94.93	98.87	90.56
ChangeMamba [17]	92.86	92.44	92.65	99.34	86.31
CDMamba [18]	96.49	89.27	92.74	99.32	86.47
TDNet(ResNet)	96.67	93.07	94.84	99.51	90.18
TDNet(PVT)	96.48	92.55	94.47	99.47	89.53
TDNet(VMamba)	97.08	92.88	94.94	99.52	90.36
TDNet(MobileSAM)	97.29	94.23	95.73	99.59	91.82

3) Comparative Experiments on WHU-CD Dataset: On the WHU-CD test set, TDNet with MobileSAM backbone achieved optimal results: 95.73% $F1$, 99.59% OA, and 91.82% IoU, as shown in Table III. Compared to A2Net (95.23% $F1$ and 90.90% IoU), this represents improvements of 0.50% in $F1$ and 0.92% in IoU.

Fig. 7 shows visual results of different CD methods on the WHU-CD dataset. Compared to LEVIR-CD and CDD, WHU-CD has higher image resolution. For Fig. 7(a), (b), and (d), most methods produce fairly accurate change maps, with errors mainly at edges. But in Fig. 7(c), due to complex backgrounds, most methods fail to detect the changed buildings. Overall, all four versions of TDNet achieve accurate CD.

4) Comparison of Efficiency and Accuracy: In this section, we evaluate the tradeoff between efficiency and accuracy across different CD methods. Our analysis shows that TDNet exhibits competitive performance across multiple metrics. As shown in Fig. 8, all variants of TDNet are distributed

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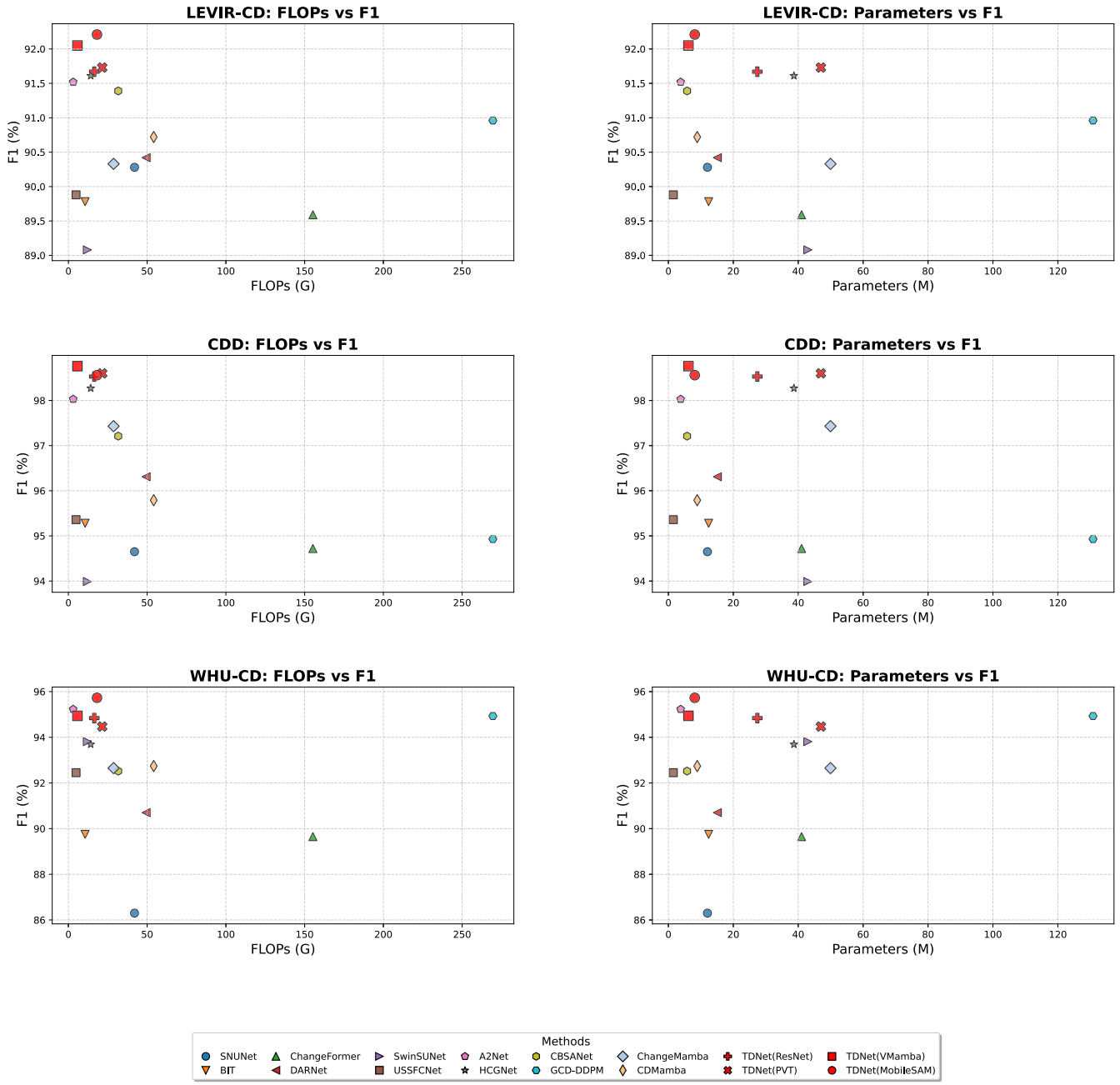


Fig. 8. Comparison of efficiency and accuracy.

in the top-left region across all datasets, demonstrating high accuracy with low complexity. Specifically, TDNet (VMamba) achieves $F1$ values of 92.05%, 98.76%, and 94.94% on the three datasets, respectively, at only 6.18M parameters and 5.75G FLOPs, yielding excellent overall performance. TDNet (MobileSAM) achieves the highest $F1$ values of 92.21% and 95.73% on the LEVIR-CD and WHU-CD datasets, respectively, while maintaining moderate complexity. Overall, TDNet achieves the optimal balance between accuracy and efficiency.

D. Ablation Studies

Ablation studies demonstrate the efficacy of the triple-decoder architecture and BLMIM, as detailed in Table IV. With the MobileSAM backbone, the full-configuration TDNet achieves 98.56% $F1$ and 97.15% IoU on CDD. Removing only the triple-decoder reduces $F1$ to 98.36% and IoU to 96.78%. Ablating BLMIM yields solely 98.41% $F1$ and 96.87% IoU. Joint removal of both components causes $F1$ and IoU to drop further to 98.09% and 96.26%, respectively. Similarly, on LEVIR-CD, the full configuration attains 92.21% $F1$ and 85.55% IoU, declining to 91.96% $F1$ and 85.12% IoU when both modules are removed. On WHU-CD, full configuration achieves 95.73% $F1$ and 91.82% IoU, while dual removal reduces performance to 95.21% $F1$ and 90.85% IoU. This demonstrates consistent stepwise degradation across all datasets.

Extended ablation across VMamba, PVT, and ResNet backbones reveals similar trends. On CDD, versus the dual components removal baseline, the full configuration delivers +0.34% $F1$ and +0.68% IoU for VMamba, +0.33% $F1$ and

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TABLE IV
ABLATION STUDY RESULTS OF TDNET WITH DIFFERENT BACKBONES

Backbone	Configuration	Params (M)	FLOPs(G)	LEVIR-CD		CDD		WHU-CD	
				F1 (%)	IoU (%)	F1 (%)	IoU (%)	F1 (%)	IoU (%)
MobileSAM	w/o Triple-Decoder + w/o BLMIM	7.29	16.07	91.96	85.12	98.09	96.26	95.21	90.85
	w/o Triple-Decoder	7.51	16.65	92.06	85.29	98.36	96.78	95.37	91.15
	w/o BLMIM	7.91	17.57	92.16	85.46	98.41	96.87	95.67	91.70
	Full configuration	8.12	18.16	92.21	85.55	98.56	97.15	95.73	91.82
	Performance gain	-	-	↑0.25	↑0.43	↑0.47	↑0.89	↑0.52	↑0.97
VMamba	w/o Triple-Decoder + w/o BLMIM	5.35	3.65	91.66	84.61	98.42	96.88	94.05	88.77
	w/o Triple-Decoder	5.56	4.24	91.82	84.88	98.56	97.17	94.55	89.66
	w/o BLMIM	5.96	5.16	91.94	85.08	98.63	97.30	94.79	90.10
	Full configuration	6.18	5.75	92.05	85.26	98.76	97.56	94.94	90.36
	Performance gain	-	-	↑0.39	↑0.65	↑0.34	↑0.68	↑0.89	↑1.59
PVT	w/o Triple-Decoder + w/o BLMIM	46.17	19.40	91.53	84.39	98.27	96.60	93.84	88.40
	w/o Triple-Decoder	46.38	19.98	91.72	84.71	98.41	96.87	94.34	89.29
	w/o BLMIM	46.78	20.91	91.71	84.68	98.48	97.00	94.30	89.21
	Full configuration	46.99	21.49	91.73	84.72	98.60	97.23	94.47	89.53
	Performance gain	-	-	↑0.20	↑0.33	↑0.33	↑0.63	↑0.63	↑1.13
ResNet	w/o Triple-Decoder + w/o BLMIM	26.57	14.36	91.38	84.14	98.05	96.17	94.29	89.20
	w/o Triple-Decoder	26.78	14.95	91.57	84.46	98.35	96.75	94.40	89.39
	w/o BLMIM	27.18	15.87	91.59	84.48	98.31	96.67	94.62	89.78
	Full configuration	27.40	16.45	91.67	84.63	98.53	97.11	94.84	90.18
	Performance gain	-	-	↑0.29	↑0.49	↑0.48	↑0.94	↑0.55	↑0.98

TABLE V
EFFICACY OF THE BLMIM

Backbone	Multi-scale interaction module	Params (M)	FLOPs(G)	LEVIR-CD			CDD		
				F1 (%)	OA (%)	IoU(%)	F1 (%)	OA (%)	IoU(%)
MobileSAM	MSCCA	7.98	17.74	91.75	99.17	84.76	98.26	99.57	96.57
	NAM	8.43	18.34	92.14	99.21	85.42	98.46	99.62	96.97
	BLMIM	8.12	18.16	92.21	99.21	85.55	98.56	99.64	97.15
ResNet	MSCCA	27.26	16.04	91.45	99.14	84.25	98.16	99.55	96.38
	NAM	28.21	16.64	91.57	99.15	84.44	98.39	99.60	96.82
	BLMIM	27.40	16.45	91.67	99.16	84.63	98.53	99.64	97.11

+0.63% IoU for PVT, and +0.48% $F1$ and +0.94% IoU for ResNet. On LEVIR-CD, gains are +0.39% $F1$ and +0.65% IoU (VMamba), +0.20% $F1$ and +0.33% IoU (PVT), and +0.29% $F1$ and +0.49% IoU (ResNet). Most notably on WHU-CD, full configuration improves over dual removal by +0.89% $F1$ and +1.59% IoU for VMamba, +0.63% $F1$ and +1.13% IoU for PVT, and +0.55% $F1$ and +0.98% IoU for ResNet.

All backbones exhibit a consistent performance ordering: *full configuration* > *single-component removal* > *dual-component removal*. This confirms that the integration of triple-decoder and BLMIM significantly enhances TDNet's representational capacity across diverse backbones.

E. Comparison of Multiscale Interaction Modules

To further validate the efficacy of BLMIM, we compare it against two other multiscale interaction modules: the multi-scale cascaded cross-attention (MSCCA) module [53] and the neighbor aggregation module (NAM) [45]. The MSCCA and the NAM enhance feature representation through the cascaded cross-attention mechanism and the local neighborhood aggregation mechanism, respectively.

Table V demonstrates the superior performance of BLMIM on both LEVIR-CD and CDD datasets across different backbones (ResNet and MobileSAM). Specifically, with the ResNet backbone, BLMIM improves the $F1$ by 0.22% and 0.12% over MSCCA and NAM, respectively, on the LEVIR-CD dataset, and by 0.37% and 0.14% on the CDD dataset. Similar improvements are observed with the MobileSAM backbone, demonstrating the effectiveness of our BLMIM.

F. Comparison of Decoder Architectures

To further validate the efficacy of the triple-decoder, we compare it against single decoder (UNet, [43]) and nested decoder (UNet++, [44]). The single decoder is the most commonly used decoder architecture in CD methods [9], [42], [45]. The nested decoder is the decoder architecture adopted in SNUNet [13]. It was proposed by UNet++ [44] and mainly enhances the performance of the network through nested, dense skip connections.

As presented in Table VI, the triple-decoder consistently improves accuracy over both single decoder and nested decoder across all evaluated backbones and datasets, validating its structural advantage.

G. Efficacy of Comprehensive Deep Supervision

To evaluate our deep supervision strategy, we conduct ablation studies from two perspectives: one expands the supervision depth, and the other expands the supervision breadth, as detailed in Tables VII and VIII, respectively.

1) Expanding Supervision in Depth: This experiment investigates the effect of progressively expanding supervision depth within the feature hierarchy.

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TABLE VI
EFFICACY OF THE TRIPLE-DECODER

Backbone	Decoder	LEVIR-CD			CDD		
		F1 (%)	OA (%)	IoU(%)	F1 (%)	OA (%)	IoU(%)
MobileSAM	Single-decoder	91.44	99.14	84.23	97.93	99.49	95.94
	Nested-decoder	91.81	99.17	84.86	98.25	99.57	96.56
	Triple-decoder	92.21	99.21	85.55	98.56	99.64	97.15
ResNet	Single-decoder	91.31	99.12	84.01	98.02	99.51	96.13
	Nested-decoder	91.35	99.13	84.08	98.38	99.60	96.81
	Triple-decoder	91.67	99.16	84.63	98.53	99.64	97.11

TABLE VII
EFFECT OF PROGRESSIVE SUPERVISION DEPTH ON MODEL PERFORMANCE

Backbone	Supervised features	LEVIR-CD			CDD		
		F1 (%)	OA (%)	IoU(%)	F1 (%)	OA (%)	IoU(%)
MobileSAM	$D_{3,1}$	91.39	99.13	84.14	98.01	99.51	96.09
	First layer	91.78	99.17	84.82	98.32	99.58	96.69
	First and second layers	91.91	99.18	85.03	98.53	99.64	97.10
	All layers	92.21	99.21	85.55	98.56	99.64	97.15
ResNet	$D_{3,1}$	90.87	99.08	83.26	98.01	99.51	96.09
	First layer	91.40	99.13	84.16	98.39	99.60	96.84
	First and second layers	91.44	99.13	84.23	98.49	99.63	97.02
	All layers	91.67	99.16	84.63	98.53	99.64	97.11

TABLE VIII
EFFECT OF PROGRESSIVE SUPERVISION BREADTH ON MODEL PERFORMANCE

Backbone	Supervised features	LEVIR-CD			CDD		
		F1 (%)	OA (%)	IoU(%)	F1 (%)	OA (%)	IoU(%)
MobileSAM	$D_{3,1}$	91.39	99.13	84.14	98.01	99.51	96.09
	Decoder D_3	91.98	99.19	85.15	98.47	99.62	96.98
	Decoder D_3 and D_2	92.11	99.20	85.37	98.54	99.64	97.11
	All decoders	92.21	99.21	85.55	98.56	99.64	97.15
ResNet	$D_{3,1}$	90.87	99.08	83.26	98.01	99.51	96.09
	Decoder D_3	91.40	99.14	84.17	98.36	99.60	96.78
	Decoder D_3 and D_2	91.51	99.15	84.35	98.46	99.62	96.96
	All decoders	91.67	99.16	84.63	98.53	99.64	97.11

1) Only the final output feature $D_{3,1}$ is supervised.
2) Supervision is extended to the first-layer outputs of all decoders ($D_{1,1}, D_{2,1}$, and $D_{3,1}$).
3) Supervision is extended to both first and second layers ($D_{1,1}, D_{2,1}, D_{3,1}, D_{1,2}, D_{2,2}$, and $D_{3,2}$).
4) Finally, supervision is extended to all decoder features.

2) Expanding Supervision in Breadth: This experiment evaluates the effect of progressively expanding the supervision breadth across the decoder cascade.

1) Only the final output feature $D_{3,1}$ is supervised.
2) Supervision is extended to all features of decoder D_3 ($D_{3,1}$–$D_{3,4}$).
3) Supervision is extended to all features of decoder D_2 and D_3 ($D_{2,1}$–$D_{2,3}$ and $D_{3,1}$–$D_{3,4}$).
4) Finally, supervision is extended to all decoder features.

The two experimental series demonstrate that expanding the supervision scope—in either depth or breadth—consistently improves $F1$ and IoU, confirming the effectiveness of our comprehensive deep supervision strategy.

H. Impact of Loss Weight Coefficients

Our loss function employs three coefficients (α_1, α_2, and α_3) to weight the losses of decoders D_1, D_2, and D_3, respectively.

TABLE IX
IMPACT OF LOSS WEIGHT COEFFICIENTS ON MODEL PERFORMANCE

$\alpha_1, \alpha_2, \alpha_3$	LEVIR-CD (%)			CDD (%)		
	F1	OA	IoU	F1	OA	IoU
$0.5, 0.5, 0.5$	92.14	99.21	85.43	98.49	99.63	97.02
$0.5, 0.5, 1$	92.21	99.21	85.54	98.57	99.65	97.18
$0.5, 1, 0.5$	92.11	99.20	85.38	98.51	99.63	97.07
$0.5, 1, 1$	92.24	99.22	85.59	98.57	99.65	97.18
$1, 0.5, 0.5$	92.02	99.20	85.22	98.47	99.62	96.98
$1, 0.5, 1$	92.23	99.21	85.57	98.54	99.64	97.13
$1, 1, 0.5$	92.14	99.20	85.43	98.49	99.63	97.03
$1, 1, 1$	92.21	99.21	85.55	98.56	99.64	97.15

To identify the optimal configuration, we conduct an ablation study using the MobileSAM backbone, evaluating eight different weight configurations. Each α_i is set to 0.5 or 1. As shown in Table IX, the configuration $(0.5, 1, 1)$ achieves the best overall performance, with $F1$ scores of 92.24% on LEVIR-CD and 98.57% on CDD.

V. CONCLUSION

In this article, we propose TDNet, a bidirectional LSTM gated TDNet, to address the insufficient exploitation of per-layer feature potential and the suboptimal multiscale interactions in existing CD methods. TDNet introduces two key

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improvements: first, a triple-decoder architecture is proposed that breaks the conventional single-decoder paradigm and employs a feature cascade refinement mechanism to progressively process shallow, middle, and deep features, thereby fully unleashing the discriminative capacity of each layer. Second, a BLMIM is designed that pioneers the adaptation of LSTM's gating mechanisms to multiscale feature fusion and allows adaptive cross-scale feature coordination. The experimental results demonstrate that TDNet achieves $F1$ improvements of 0.6%, 0.49%, and 0.5% over state-of-the-art methods on LEVIR-CD, CDD, and WHU-CD datasets, respectively.

Although the triple-decoder architecture seems like a simple structural modification, its design is aimed at enhancing the discriminative capability of each layer. This architecture boosts network performance and establishes a novel architectural paradigm for future methods. As the pioneering module leveraging LSTM for supervised CD, BLMIM provides a novel perspective for future approaches. We will release the code so others can build on these two components straight away.

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